

NLP Problem Statement – ASSIGNMENT PS-5

Title

Intelligent News Article Classification and Fake News Detection using NLP

Problem Statement

With the rapid growth of online media platforms, large volumes of news articles are published every day. Identifying fake news and classifying news articles manually is difficult and time-consuming. An intelligent NLP-based system can automate news analysis, extract important information, understand contextual meaning, and classify articles into appropriate news categories while detecting misleading content.

In this project, students are required to design and implement an NLP pipeline using a real-world news dataset to automatically analyze news articles and classify them into different categories.

Suggested Dataset

- Fake News Dataset (Kaggle)
- News Category Dataset
- Real and Fake News Dataset

The developed system should preprocess news article text and perform the following NLP tasks:

1. Neural Language Model (Neural LM):

Develop a Neural Language Model to understand news article language patterns and predict contextual words related to news content.

2. Vector Embedding:

Generate vector representations of news text using Word2Vec, GloVe, or FastText for semantic understanding and similarity analysis.

3. POS Tagging:

Implement Part-of-Speech tagging to identify named entities, action verbs, political terms, and important keywords.

4. Hidden Markov Model (HMM):

Use HMM-based probabilistic sequence modeling for news tagging and fake news detection tasks.

5. Parsing:

Perform syntactic parsing to analyze sentence structure and extract meaningful relationships between entities, events, and claims.

The system should classify news articles into categories such as:

- Politics
- Technology
- Sports
- Entertainment
- Business
- Health
- Fake News / Real News

Final Output

The system should display:

- Extracted keywords and entities
- Predicted news category
- Fake news detection result
- NLP analysis results
- Performance metrics

Bonus Marks

Award: 1 Mark Extra Credit will be given for successful implementation using a Virtual Lab.